

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour
Total Marks: 30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

(192573016)

I N. PRAANVA VENKATESH (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



(88)

88

D) FLIGHT DELAY NOTIFICATION

1. Propositional logic

* D = Flight 202 is delayed.

* N = Passengers are notified

Rule : $D \rightarrow N$

Fact : D

Goal : N

2. CNF and Negated Goal

Implication:

$$D \rightarrow N \equiv \neg D \vee N$$

Clauses:

* $C_1 : \neg D \vee N$

* $C_2 : D$

* Negated goal : $C_3 : \neg N$

3. Resolution

$C_1 + C_2 :$

$$(\neg D \vee N) + D \rightarrow N$$

Then:

$$N + \neg N \rightarrow \square$$

4. Conclusion.

Since the Empty Clause ($\square$) is derived, the goal is proved.

$\therefore$ Passengers are notified. $\checkmark$

2) Employee Certification

1. Propositional Logic

* T = Kevin completed training.

* C = Kevin receives certification.

Rule: $T \rightarrow C$

Fact: T

Goal: C

Assumes: CNF and Negated Goal.

* $C1: \neg T \vee C$

$$T \rightarrow C \equiv \neg T \vee C$$

* $C2: T$

* $C3: \neg C$

3. Resolution:

$$C1 + C2$$

$$(\neg T \vee C) + T \rightarrow C$$

$$C + \neg C:$$

$$C + \neg C \rightarrow \square$$

4. Conclusion

$\square$ is obtained.

$\therefore$ Kevin receives certification.

Library Late Fine

1. Propositional Logic

* L = Book was returned late.

* F = Fine is charged.

Rule : $L \rightarrow F$

Fact : L

Goal : F

2. CNF and Negated Goal

$$L \rightarrow F \equiv \neg L \vee F$$

Clauses:

* $C1 : \neg L \vee F$

* $C2 : L$

* $C3 : \neg F$

3. Resolution

$$C1 + C2:$$

$$(\neg L \vee F) + L \rightarrow F$$

$$F + \neg F:$$

$$F + \neg F \rightarrow \square$$

4. Conclusion

The Empty Clause $\square$ is derived.

$\therefore$ A fine is charged.

3) Medical Attention Alert

1. Propositional Logic

* F = Ravi has fever

* M = Ravi needs medical attention.

Rule: $F \rightarrow M$

Fact: F

Goal: M

2. CNF and Negated Goal

$$F \rightarrow M \equiv \neg F \vee M$$

Clauses:

$$* C1: \neg F \vee M$$

$$* C2: F$$

$$* C3: \neg M$$

3. Resolution

$$C1 + C2$$

$$(\neg F \vee M) + F \rightarrow M$$

$$M + \neg M:$$

$$M + \neg M \rightarrow \square$$

4. Conclusion

$\square$ proves the goal

$\therefore$ Ravi needs medical attention.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks (100)
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	22
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	22
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	18
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	18
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	8

(88)

10